

Omar Kayali

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EDUCATION

Bachelor of Science, Data Science, May 2027

College of Science and Engineering, University of Minnesota-Twin Cities, Minneapolis, MN

GPA: 3.51

PROJECTS

Life Expectancy Classification, January 2026 – Current

Applied Machine Learning

- Developed a nearest neighbors classifier using WHO data of about 20 metrics to predict whether a country achieves life expectancy of at least 75 years
- Split and normalized data to reduce noise, improve model performance, and prevent data leakage and overfitting
- Produced pairplots between features using Pandas to inform feature selection

PostgreSQL Page Replacement in C, January 2026 – Current

Architecture and Implementation of Database Management Systems

- Implemented an LRU page replacement policy within the PostgreSQL buffer manager to control memory eviction behavior
- Used GDB and Valgrind in a virtual Docker environment to prevent memory leaks, ensure correctness, and test under simulated workloads to verify stability

Laptop Price Prediction & Brand Classifier, January 2026 – Current

Applied Machine Learning

- Built two PyTorch linear regression models to predict laptop prices from hardware features, improving R^2 by training on a broader feature set
- Trained a logistic regression classifier in PyTorch to identify Apple laptops, utilizing class weighting inversely proportional to label frequency to address an imbalanced dataset
- Tuned classification threshold from 0.5 to 0.75 by iterating across values and maximizing F1 score; computed precision, recall, F1 and MCC on testing data

SKILLS

Programming: Python, R, SQL, Java, C, Git, PyTorch, HTML, CSS

Statistical Analysis: Regression, Classification, Visualization

Mathematics: Linear Algebra, Multivariable Calculus, Statistics

WORK EXPERIENCE

Student Website Manager, September 2023 – Current

University of Minnesota Alumni Association, Minneapolis, MN

- Edits HTML and formats images to add and update website content daily
- Adapts *Minnesota Alumni* magazine to an accessible digital format
- Uses accessibility data to ensure website is ADA compliant
- Creates mass emails through Salesforce to inform members of UMAA events

Tutor, September 2024 – January 2025

Varsity Tutors

- Digitally tutored high school students in calculus, trigonometry, algebra, and ACT/SAT preparation
- Worked closely in a one on one live environment with students to best accommodate each one's learning style
- Created and graded practice problems to provide students constructive feedback through